B.M.S. College of Engineering

(Autonomous Institution affiliated to VTU, Belagavi)

Department of Computer Science and Engineering

AAT Verilog Laboratory Report 23CS3PCLOD

(September 2025-January 2026)

B.M.S. College of Engineering

Department of Computer Science and Engineering

Laboratory Certificate

This is to certify that Himani.P, Keerthi M patil, Harshitha.M
has satisfactorily completed the course of Experiments in Practical Logic Design (Verilog) prescribed by the
Department during the odd semester 2025-26.
Name of the Candidate: Himani.P, Keerthi M patil, Harshitha.M

USN No.: **1WA24CS129, 1WA24CS146, 1WA24CS127** Semester: **III** Section: **N**

Marks	
Max. Marks	Obtained

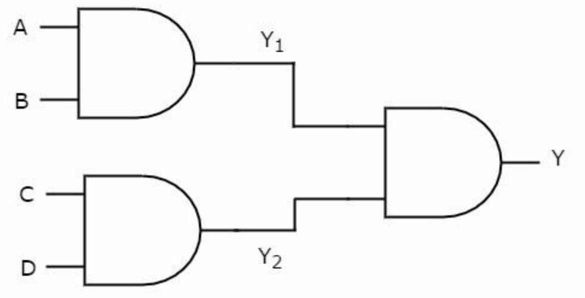
10	
Marks in Words	

Signature of the staff in-charge **Head of the Department**

Date: 3-01-2026

Verilog Program List
23CS3PCLOD
Laboratory Experiments

Serial No.	Title	Page No
	CYCLE I **Structural Modelling**	
1.	Write HDL implementation for the following Logic a. AND/OR/NOT Simulate the same using structural model and depict the timing diagram for valid inputs.	
1.	Write HDL implementation for the following Logic a. NAND/NOR Simulate the same using structural model and depict the timing diagram for valid inputs.	

1.	Write HDL implementation for the following AND-OR Combinational Logic  Simulate the same using structural model and depict the timing diagram for valid inputs.	
1.	Write HDL implementation for a 4:1 Multiplexer. Simulate the same using structural model and depict the timing diagram for valid inputs.	
1.	Write HDL implementation for a 2-to-4 decoder. Simulate the same using structural model and depict the timing diagram for valid inputs.	
6.	Write HDL implementation for a 4-to-2 encoder. Simulate the same using structural model and depict the timing diagram for valid inputs.	
	CYCLE II **Behavior Modeling**	
1.	Write HDL implementation for a RS flip-flop using behavioral model. Simulate the same using structural model and depict the	

	timing diagram for valid inputs.	
1.	Write HDL implementation for a JK flip-flop using behavioral model. Simulate the same using structural model and depict the timing diagram for valid inputs.	
9.	Write HDL implementation for a 3-bit up-counter using behavioral model. Simulate the same using structural model and depict the timing diagram for valid inputs.	
	CYCLE III **Dataflow Modeling**	
10.	Write HDL implementation for AND/OR/NOT gates using data flow model. Simulate the same using structural model and depict the timing diagram for valid inputs	

Verilog Program List-23CS3PCLOD

SCHEME OF CONDUCT AND EVALUATION

CLASS: III SEMESTER
YEAR: 25-26

EVALUATION SCHEME Tutorial Test: 1 hour

Expt. No.	TITLE	Max. Marks	Marks Obtained	Signature
1.	AND/OR/NOT			
1.	NAND/NOR			
1.	Logic diagram			
1.	Multiplexer			
1.	Decoder	5		
1.	Encoder			
1.	RS			
1.	JK			
1.	Counter			
1.	AND/OR/NOT – data flow			
	Test: Viva – 2 Marks + Writeup – 1 Mark + Execution – 2 Marks	5		
	TOTAL MARKS	10		

Icarus Verilog Installation (Windows)

1. Download and install iverilog from here: http://bleyer.org/icarus/
2. During installation, check the checkbox so that iverilog is added to the System PATH.
3. If this is not done, one will need to manually locate their iverilog installation directory and copy the path to the bin folder situated within it.
4. To do this, we will need to navigate to the control panel and access the environment variables section of the computer. Here, we add a new variable and set it to `%PATH%;c:\iverilog\bin` assuming that is the installation directory for iverilog.
5. Open command prompt or any other preferred terminal and type `iverilog` and press enter. Trace back the steps for mistakes if version information is not displayed and re-install if necessary.

Compilation and viewing waveforms

1. With iverilog, a third party text editor is necessary. Any editor would work (notepad, gedit, or vim, for example) but a modern text editor such as SublimeText 3 or Visual Studio Code is preferred as plug-ins can be installed to provide syntax highlighting for the Verilog HDL code being written.
2. While writing the code, two lines must be added in the test bench module.

```
module testSR;
    reg [1:0]A;
    reg c;
    wire q,qb;
    SR_FF srff(A,c,q,qb);
    initial c = 1'b0;
    always #5 c = ~c;
    initial
        begin
            $dumpfile("test.vcd");
            $dumpvars(0,testSR);
```

   The $dumpfile() and $dumpvars() functions should be passed, with the former containing the name of a file ending with .vcd and the latter containing the name of the testbench module itself. This will dump a vcd file that will allow us to view the waveform. This should always be right after the initial and begin statements.
3. After this file is written and saved under a .v extension (example `VHDLcode.v`), the terminal must be opened and the user must navigate to the directory in which the file is saved.
4. In this directory, running `iverilog -o outputFile VHDLcode.v` will output a .vvp file with the name `outputFile.vvp`.
5. After generating the vvp file, run the `vvp outputFile` command to get the waveform dump with name of the given string under $dumpfile().
6. The next step is to run `gtkwave` in the command line and open File > Open New Tab > select the generated .vcd file. Afterward, click on the added element on the viewer and insert it. This will display the waveform.

References

1. Zucker, M. (2019). *E15 - Installing and testing Icarus Verilog*. [online] Swarthmore.edu. Available at: https://www.swarthmore.edu/NatSci/mzucker1/e15_f2014/iverilog.html
2. KONSTADELIAS, I. (n.d.). *Icarus Verilog + GTKWave Guide*. [ebook] Available at: http://inf-server.inf.uth.gr/~konstadel/resources/Icarus_Verilog_GTKWave_guide.pdf

CYCLE 1: STRUCTURAL MODELLING

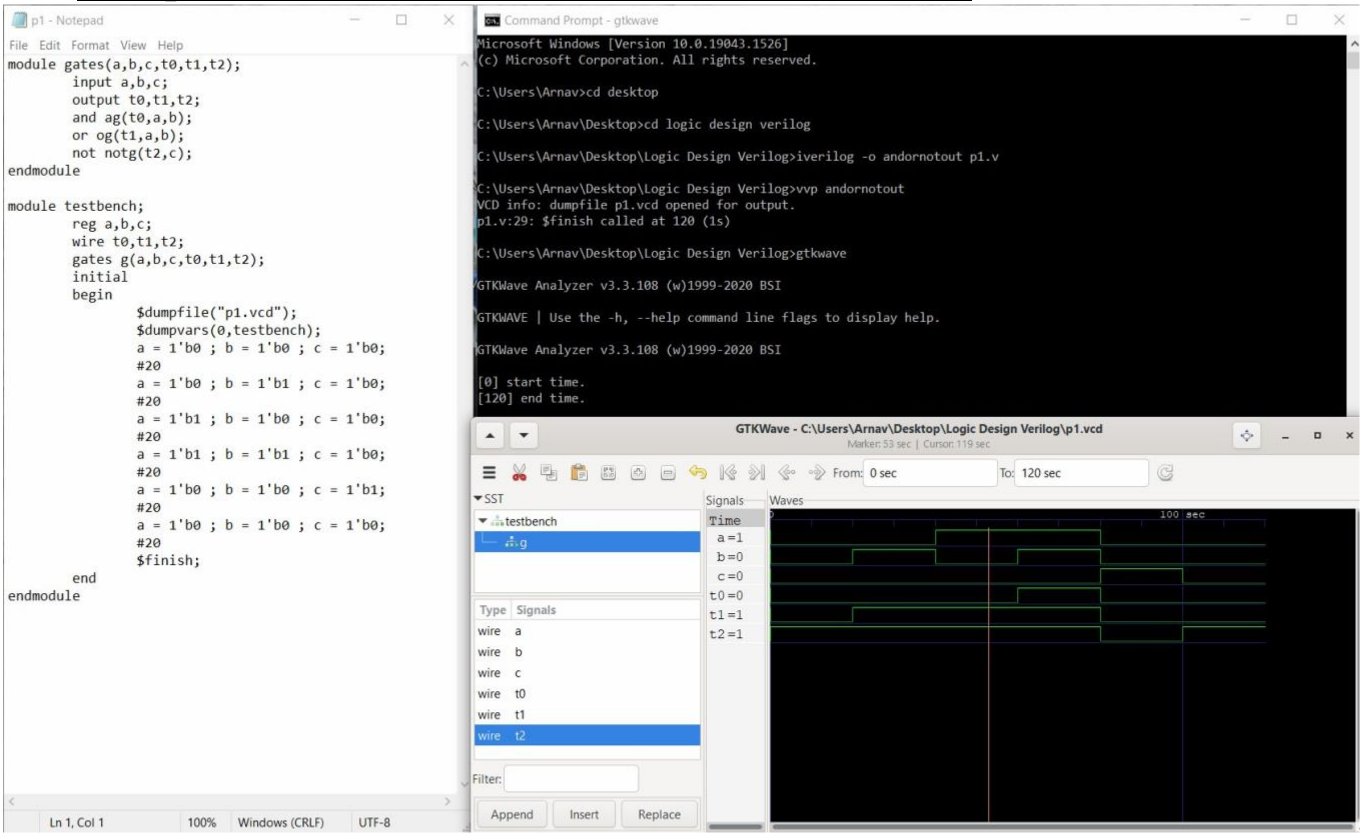
Experiment 1

```
module gates(a,b,c,t0,t1,t2);     // start of module gates with following variables
    input a,b,c;          // the inputs to the given circuit are a,b,c
```

```
        output t0,t1,t2;        //the outputs from the gates are t0,t1,t2
        and ag(t0,a,b);         // and gate ag takes input a,b and gives output t0
        or og(t1,a,b);          // or gate og takes input a,b and gives output t1
        not notg(t2,c);         // not gate notg takes c as input and gives output t2
endmodule                       //end of the module

module testbench;               //start of testbench
        reg a,b,c;   //a,b,c are data storage elements , synthesized to combinational circuit
        wire t0,t1,t2;   //t0,t1,t2 are assigned to be connected gates
        g(a,b,c,t0,t1,t2);
        initial //initial processes execute only once begin
           $dumpfile("gates.vcd"vvp ); //Level set to 0 implies that all variables of
module
           $dumpvars(0,testbench);  into the gates.vcd file(to show output in gtkwave)
        a = 1'b0 ; b= 1'b0 ; c= 1'b0;    //for the first 20ns inputs a,b,c are given
values
           #20                              0,0,1 represented by a single bit
             a = 1'b0 ; b= 1'b1 ; c= 1'b0;          //for the next 20ns inputs a,b,c are given
values
           #20                              0,1,0(single bit representation)
           a = 1'b1 ; b= 1'b0 ; c= 1'b0;     //for next 20ns a,b,c have the values 1,0,0
        #20
           a = 1'b1 ; b= 1'b1 ; c= 1'b0;     //for next 20ns a,b,c have values 1,1,0
        #20
           a = 1'b0 ; b= 1'b0 ; c= 1'b1;     //for next 20ns a,b,c have values 0,0,1
        #20
           a = 1'b0 ; b= 1'b0 ; c= 1'b0;     //for next 20ns a,b,c have values 0,0,0
        #20
           $finish;                         //$finish denotes the end of time duration for
        End                                  which input values are given to a,b,c
endmodule
```

Compilation, Execution and Result of Simulation



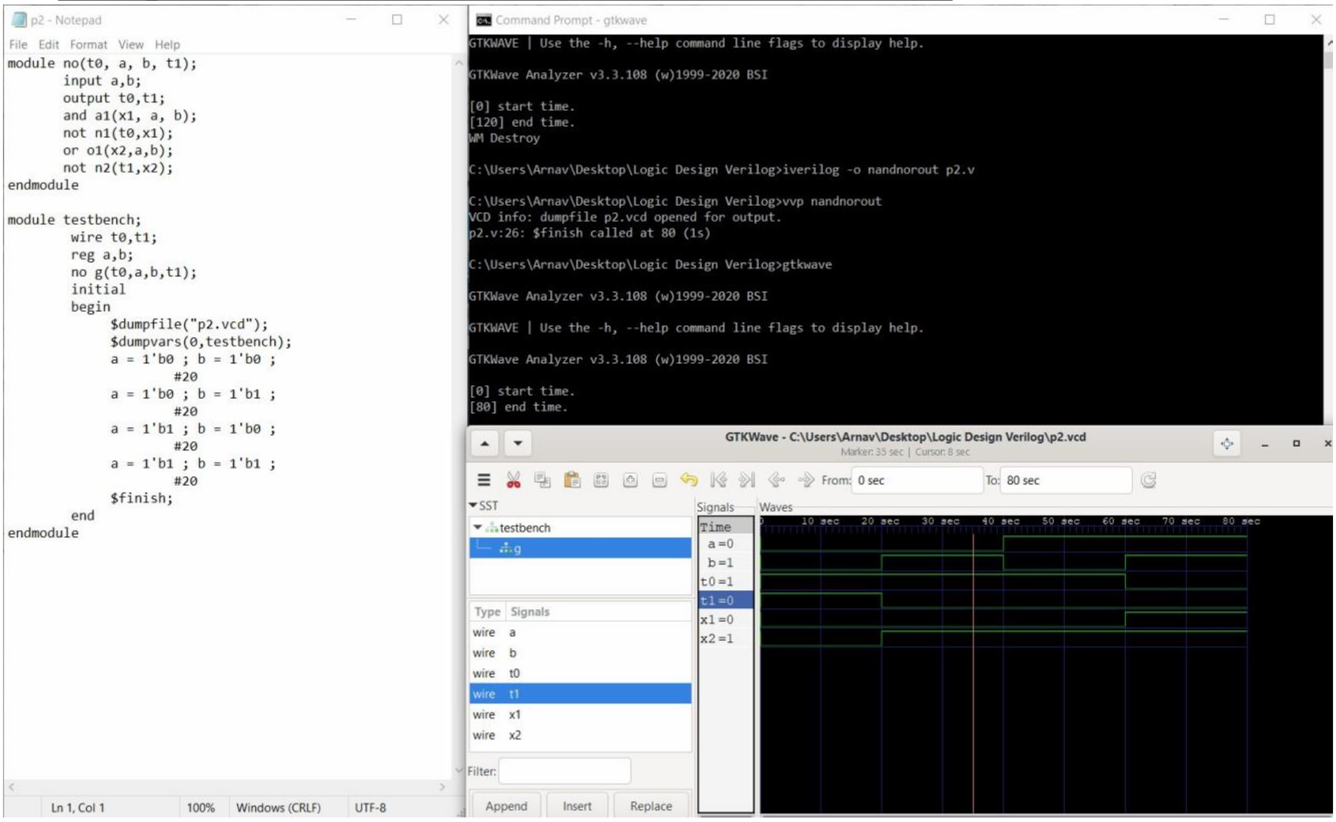
Experiment 2

```
module no(t0,a,b,t1);          // start of module no with following variables
        input a,b;             //a and b are inputs for the circuit
        output t0,t1;          //t0 and t1 are outputs for the circuit
        and a1(x1,a,b);        //a1 represents and gate with a,b inputs and x1 output
        not n1(t0,x1);         //n1 represents not gate with x1 input and t0 output
        or o1(x2,a,b);         //o1 represents or gate with a,b inputs and x2 output
        not n2(t1,x2);         //n2 represents not gate with x2 input and t1 output
endmodule              // x1,x2 are intermediate inputs/output for respective gates

module testbench;
        wire t0,t1; reg
        a,b; no
        g(t0,a,b,t1);
        initial                //initial processes execute only once
        begin          $dumpfile("gates12.vcd");//All variables of module dumped to
gates12.vcd
                $dumpvars(0,testbench);
```

```
        a = 1'b0 ; b=1'b0;          //for the first 20ns inputs a,b have values 0,0
    #20
        a = 1'b0 ; b=1'b1;          //for the next 20ns inputs a,b have values 0,1
    #20
        a = 1'b1 ; b=1'b0;          //for the next 20ns inputs a,b have values 1,0
    #20
        a = 1'b1 ; b=1'b1;          //for the next 20ns inputs a,b have values 1,1
    #20
        $finish;                    //$finish denotes the end of time duration for
    End                                which input values are given to a and b
endmodule
```

Compilation, Execution and Result of Simulation



Experiment 3

```
module andor(A,B,C,D,Y); input A,B,C,D;        //inputs for the circuit are
    a,b,c,d output Y;      //y is the output of the circuit wire
    and_op1,and_op2; and g1(and_op1,A,B);      //g1 represents an
    and gate and g2(and_op2,C,D);     //g2 represents an and gate or
    g3(Y,and_op1,and_op2);            //g3 represents the or gate
endmodule
```

module test_andor;

reg a,b,c,d; //a,b,c,d treated as data storage elements

wire y; andor ao(a,b,c,d,y); initial //initial processes execute only once begin

$dumpfile("diagram_test.vcd");

$dumpvars(0,test_andor);

a=0;b=1;c=1;d=1; //for the first 10ns a,b,c,d have the values 0,1,1,1

#10

a=0;b=0;c=1;d=0; //for the next 10ns a,b,c,d have the values 0,0,1,0

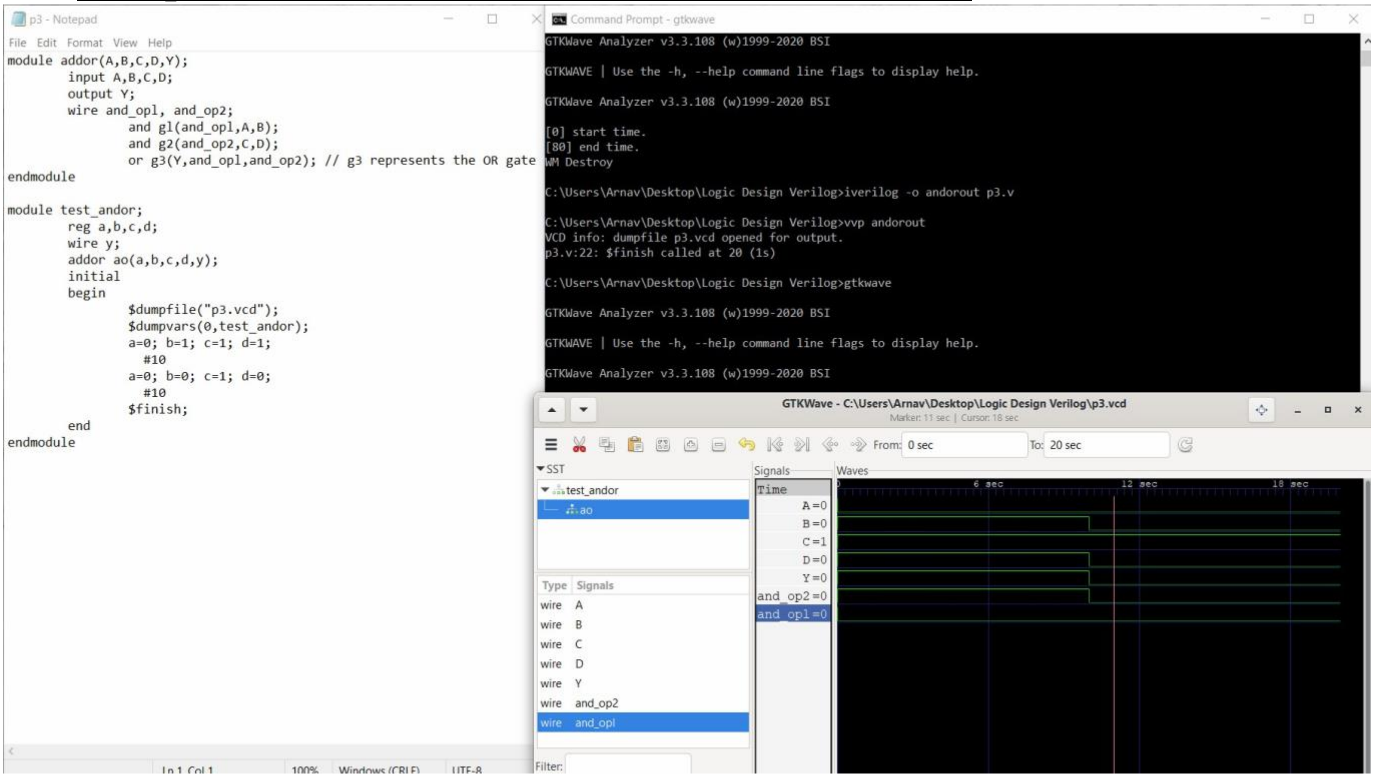
#10

$finish; //$finish denotes the end of time duration for which input values are given to a,b,c

end

endmodule

Compilation, Execution and Result of Simulation



Experiment 4

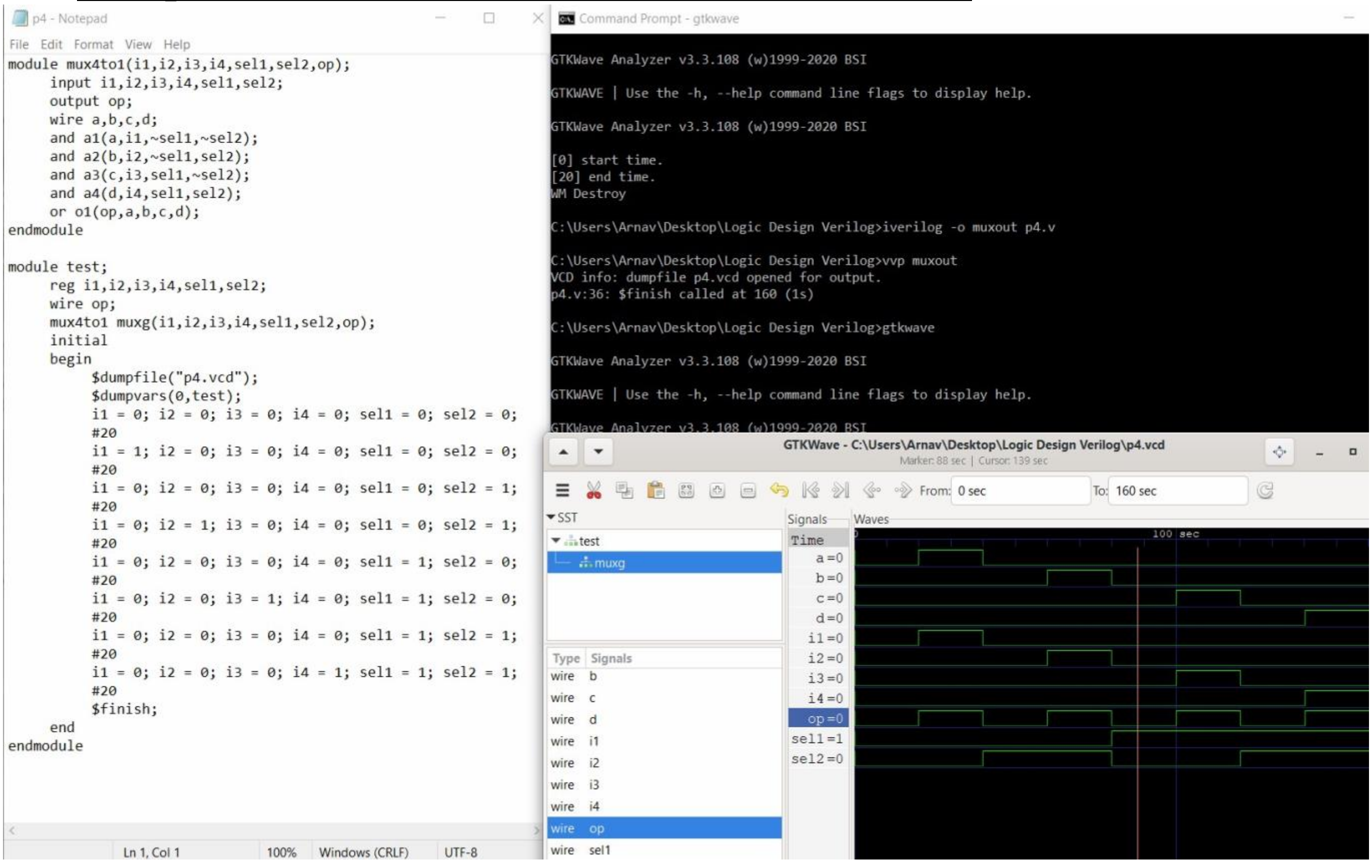
module Mux4to1(t1,t2,t3,t4, sel1, sel2,op); //Mux4to1 module declaration with all

input t1,t2,t3,t4,sel1,sel2; //involved variables output op; wire a,b,c,d; and a1(a,t1,-sel1,-sel2); //a1 is an and gate with a as output, t1,-sel1,-sel2 inputs and a2(b,t2,-sel1,sel2); //Similarly a2,a3,a4 are AND gates with respective

```
        and a3(c,t3,sel1,-sel2);     inputs and outputs and
a4(d,t4,sel1,sel2);     or o1(op,a,b,c,d);          //o1 is an or gate
with op as output, endmodule

module test; reg
        t1,t2,t3,t4,sel1,sel2;
        wire op;
        Mux4to1 muxg(t1,t2,t3,t4,sel1,sel2,op);
        initial begin
                $dumpfile("muxout.vcd");
                $dumpvars(0,test);
                t1=0;t2=0;t3=0;t4=0;sel1=0;sel2=0;     //inputs t1,t2,t3,t4 are given
values
                    #20          0,0,0,0 for the first 20ns and sel1=0,sel2=0
        t1=1;t2=0;t3=0;t4=0;sel1=0;sel2=0;    //inputs t1,t2,t3,t4 are given values
                 #20                                  1,0,0,0 for the next 20ns and
sel1=0,sel2=0
                t1=0;t2=0;t3=0;t4=0;sel1=0;sel2=1;   //inputs t1,t2,t3,t4 are given values
                 #20                                  0,0,0,0 for the next 20ns and
sel1=0,sel2=1
                t1=1;t2=1;t3=0;t4=0;sel1=0;sel2=1;   //inputs t1,t2,t3,t4 are given values
                 #20                                  0,0,0,0 for the next 20ns and
sel1=1,sel2=1
                t1=0;t2=0;t3=1;t4=0;sel1=1;sel2=0;   //inputs t1,t2,t3,t4 are given values
                 #20                                  0,0,0,0 for the next 20ns and
sel1=1,sel2=0
                t1=1;t2=1;t3=0;t4=1;sel1=1;sel2=0;   //inputs t1,t2,t3,t4 are given values
                 #20                                  0,0,0,0 for the next 20ns and
sel1=1,sel2=0
                t1=1;t2=0;t3=0;t4=1;sel1=1;sel2=1;   //inputs t1,t2,t3,t4 are given values
                 #20                                  0,0,0,0 for the next 20ns and
sel1=1,sel2=1 t1=0;t2=1;t3=1;t4=0;sel1=1;sel2=1;   //inputs t1,t2,t3,t4 are given values
                #20                            0,0,0,0 for the next 20ns and
sel1=1,sel2=1
                $finish;                        // $finish denotes the time duration for
which
        end                                     inputs t1,t2,t3,t4 and select lines sel1,sel2
endmodule
```

Compilation, Execution and Result of Simulation



Experiment 5

module bcddecoder2to4(b0,b1,d0,d1,d2,d3);

input b0,b1; //inputs b0,b1 and outputs d0,d1,d2 and d3 output d0,d1,d2,d3; wire t0,t1; not n1(t0,b0);//n1,n2 represent the NOT gates not n2(t1,b1); and a0(d0,t0,t1); //a0 represents an and gate , d0 output and t0,t1,inputs and a1(d1,t0,b1); //Similarly a1,a2,a3 represent the other AND gates and a2(d2,b0,t1); and a3(d3,b0,t1); endmodule

module test; reg b0,b1;

wire d0,d1,d2,d3;

bcddecoder2to4 bcdg(b0,b1,d0,d1,d2,d3);

initial begin

$dumpfile("bcd.vcd");

$dumpvars(0,test);

b0 = 0; b1 = 0; //for the first 40ns b0,b1 have values 0,0

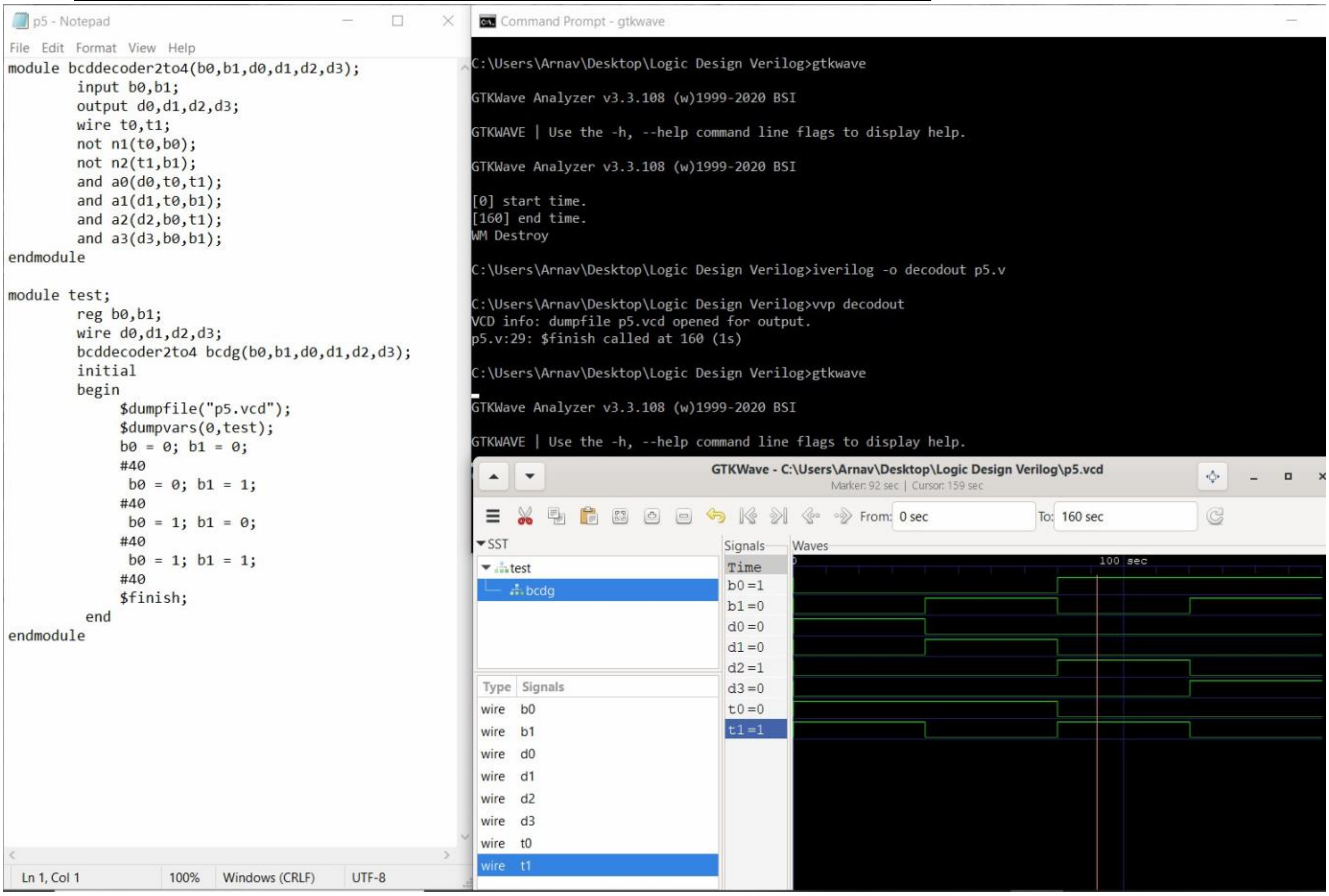
#40

b0 = 0; b1 = 1; //for the next 40ns b0,b1 have values 0,1

#40

```
            b0 = 1; b1 = 0;                    //for the next 40ns b0,b1 have values 1,0
            #40
            b0 = 1; b1 = 1;                    //for the next 40ns b0,b1 have values 1,1
            #40
            $finish;
      end
endmodule
```

Compilation, Execution and Result of Simulation



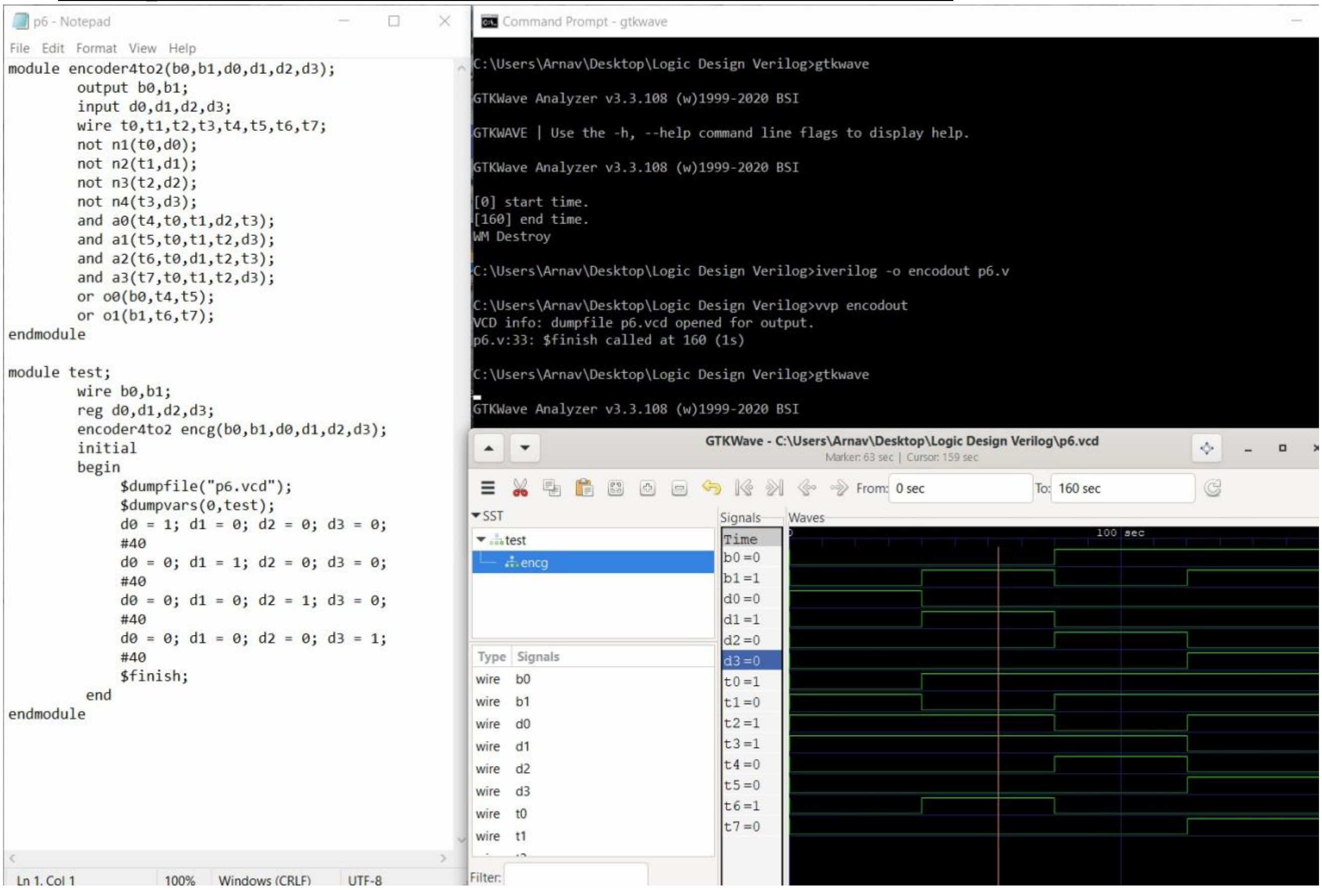
Experiment 6

module encoder4to2(b0,b1,d0,d1,d2,d3); output b0,b1; input d0,d1,d2,d3; not n0(t0,d0); //n0,n1,n2.n3 represent the NOT

```
gates not n1(t1,d1); not n2(t2,d2); not n3(t3,d3); and
a0(t4,t0,t1,d2,t3);  //a0,a1,a2,a3 represent the AND gates and
a1(t5,t0,t1,t2,d3); and a2(t6,t0,d1,t2,t3); and a3(t7,t0,t1,t2,t3);
or o0(b0,t4,t3);      //o0,o1 represent the OR gates or o1(b1,t6,d3);
endmodule

module test; wire
        b0,b1; reg
        d0,d1,d2,d3;
        encoder4to2 encg(b0,b1,d0,d1,d2,d3);
initial         begin
                $dumpfile("enc4to2.vcd");
                $dumpvars(0,test);
                d0 = 1;d1 = 0;d2 = 0;d3 = 0;      //for the first 40ns inputs d0,d1,d2,d3 have
                #40;                                the values 0,0,0,0
                d0 = 0;d1 = 1;d2 = 0;d3 = 0;      //for the next 40ns inputs d0,d1,d2,d3 have
                #40;                                the values 0,1,0,0
                d0 = 0;d1 = 0;d2 = 1;d3 = 0;      //for the next 40ns inputs d0,d1,d2,d3 have
                #40;                                the values 0,0,1,0
                d0 = 0;d1 = 0;d2 = 0;d3 = 1;      //for the next 40ns inputs d0,d1,d2,d3 have
                #40;   the values 0,0,0,1 end
endmodule
```

Compilation, Execution and Result of Simulation



CYCLE 2: BEHAVIOR MODELLING

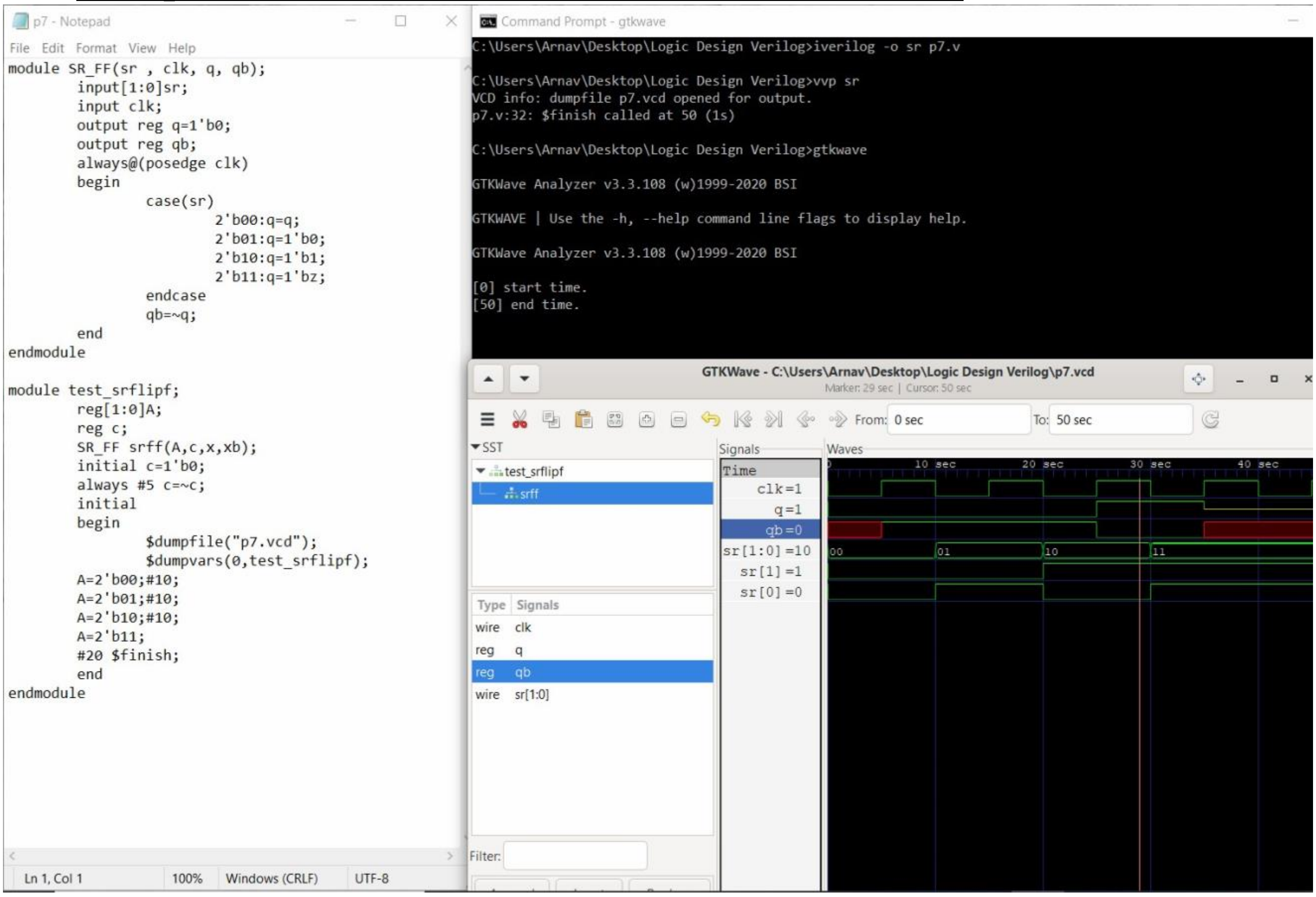
Experiment 7

```
module SR_FF(sr , clk, q, qb); input[1:0]sr; //[1:0]sr represents the
        RS Flip-flop input input clk; //clk represents the clock of the
        circuit
        output reg q=1'b0; //q and qb represent data storage elements synthesized output
        reg qb; //for sequential logic circuit
        always@(posedge clk)    //code inside block executed at every positive edge
        begin case(sr)
                2'b00:q=q;      //statements which demonstrate the flip-flops behavior
                2'b01:q=1'b0;       //based on the inputs
                2'b10:q=1'b1;
                2'b11:q=1'bz;       //forbidden state
            endcase qb=~q;

        end
endmodule
```

```
module test_srflipf;
        reg[1:0]A;
        reg c;
        SR_FF srff(A,c,x,xb);
        initial c=1'b0;
        always #5 c=~c;
        initial
        begin
        $dumpfile(
        "sr_test1.v
        cd");
                $dumpvars(0,test_srflipf);
        A=2'b00;#10;     //For first 10ns, value of A is 00, in the next 10ns the value is 01
        A=2'b01;#10;        //Then 10 and later 11.(This stage is allowed for 20ns)
        A=2'b10;#10;
        A=2'b11;
        #20 $finish; //$finish denotes end of the time duration for which values are given
        end
endmodule
```

Compilation, Execution and Result of Simulation



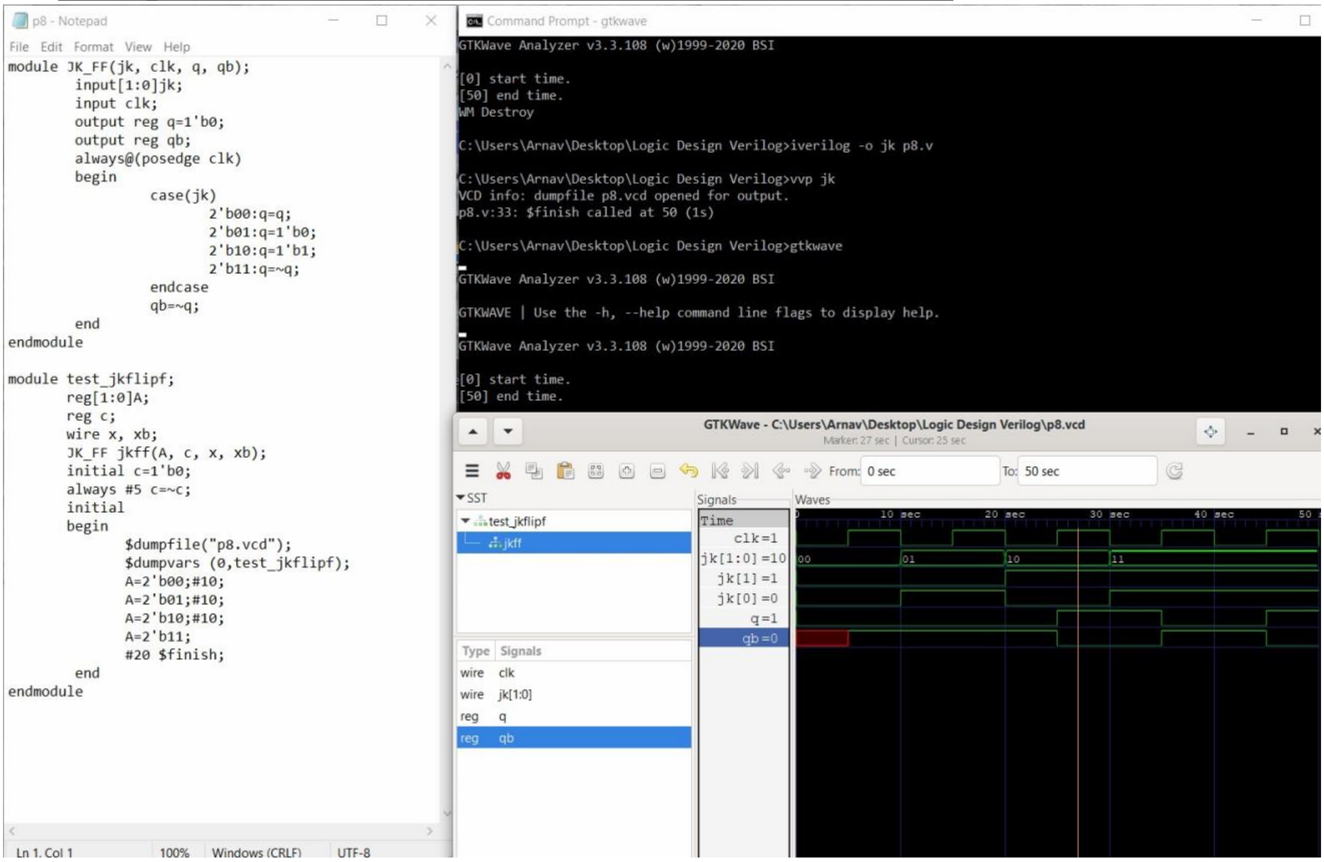
Experiment 8

```
module JK_FF(jk , clk, q, qb); input[1:0]jk; input clk; output reg
        q=1'b0; //storage elemnts synthesized for sequential ciruit output reg
        qb; always@(posedge clk) begin case(jk)
                        2'b00:q=q;          //Q(n+1) = Qn , 0 , 1, (Qn)'
                        2'b01:q=1'b0;
                        2'b10:q=1'b1;
                        2'b11:q=~q;
                endcase qb=~q;
        end
endmodule

module test_jkflipf; reg[1:0]A; reg c; wire x,xb; JK_FF
        srff(A,c,x,xb); initial c=1'b0; always #5 c=~c;   //for every
        5ns c is complemented forever initial begin
                $dumpfile("jk_test1.vcd"); //all variables of module dumped to
jk_test1.vcd
                $dumpvars(0,test_jkflipf);
```

```
        A=2'b00;#10;
        A=2'b01;#10;
        A=2'b10;#10;
        A=2'b11;
        #20 $finish;
        end
endmodule
```

Compilation, Execution and Result of Simulation

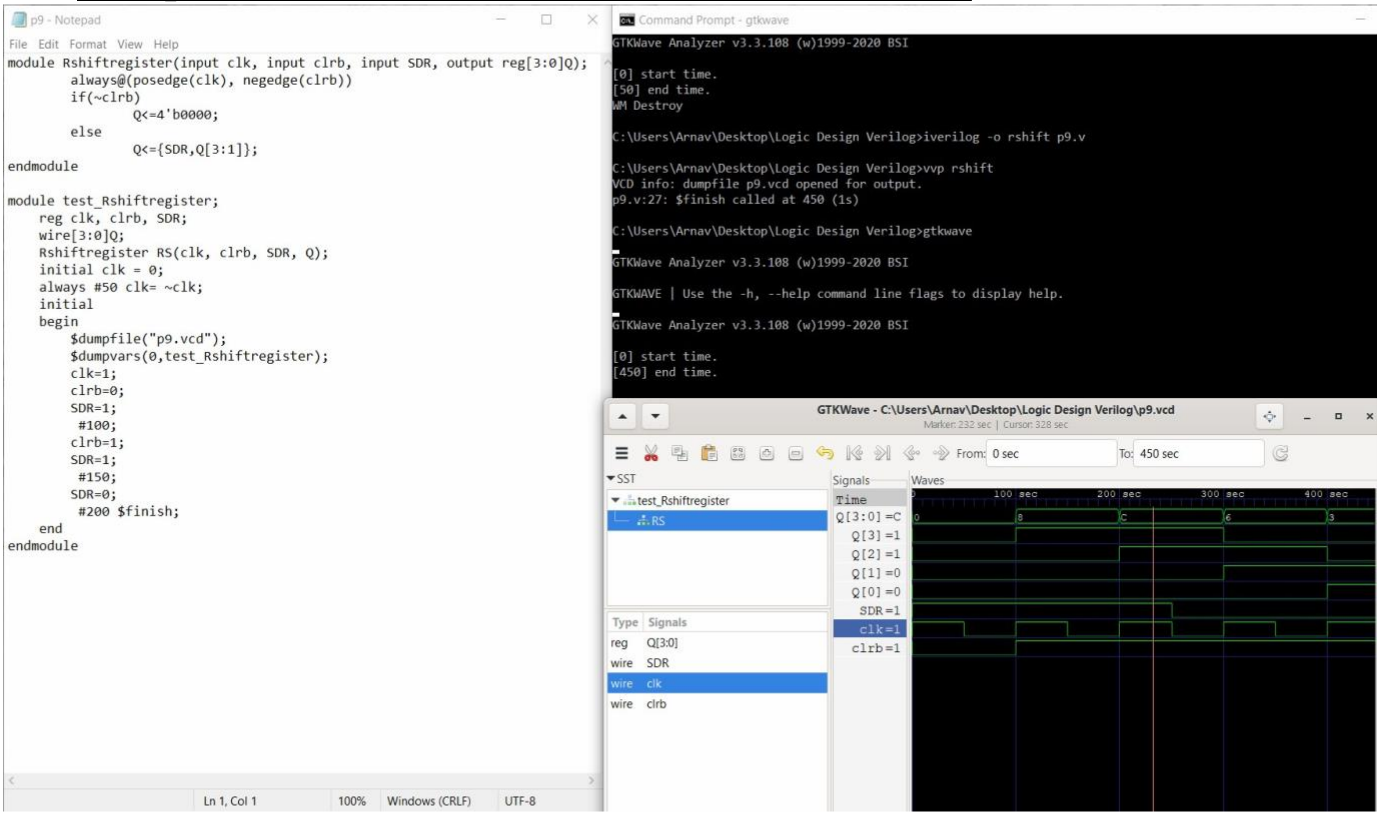


Experiment 9

```
module Rshiftregister(input clk, input clrb, input SDR, output reg[3:0]Q);
        always@(posedge (clk),negedge(clrb)) if(~clrb)
                Q<=4'b0000;  //4-bit shift register
        else
                Q<={SDR,Q[3:1]};
endmodule
```

```
module test_Rshiftregister;
        reg clk,clrb,SDR;
        wire[3:0]Q;
        Rshiftregister RS(clk,clrb,SDR,Q); initial clk = 0;
        always #50 clk=~clk;  //for every 50ns clk is complemented
        initial begin
                $dumpfile("test_Shreg1.vcd");
                $dumpvars(0,test_Rshiftregister);
                clk=1; clrb=0; SDR = 1;
                  #100;
                clrb=1;
                SDR = 1;
                  #150;
                SDR=0;
                  #200 $finish;
        end
endmodule
```

Compilation, Execution and Result of Simulation



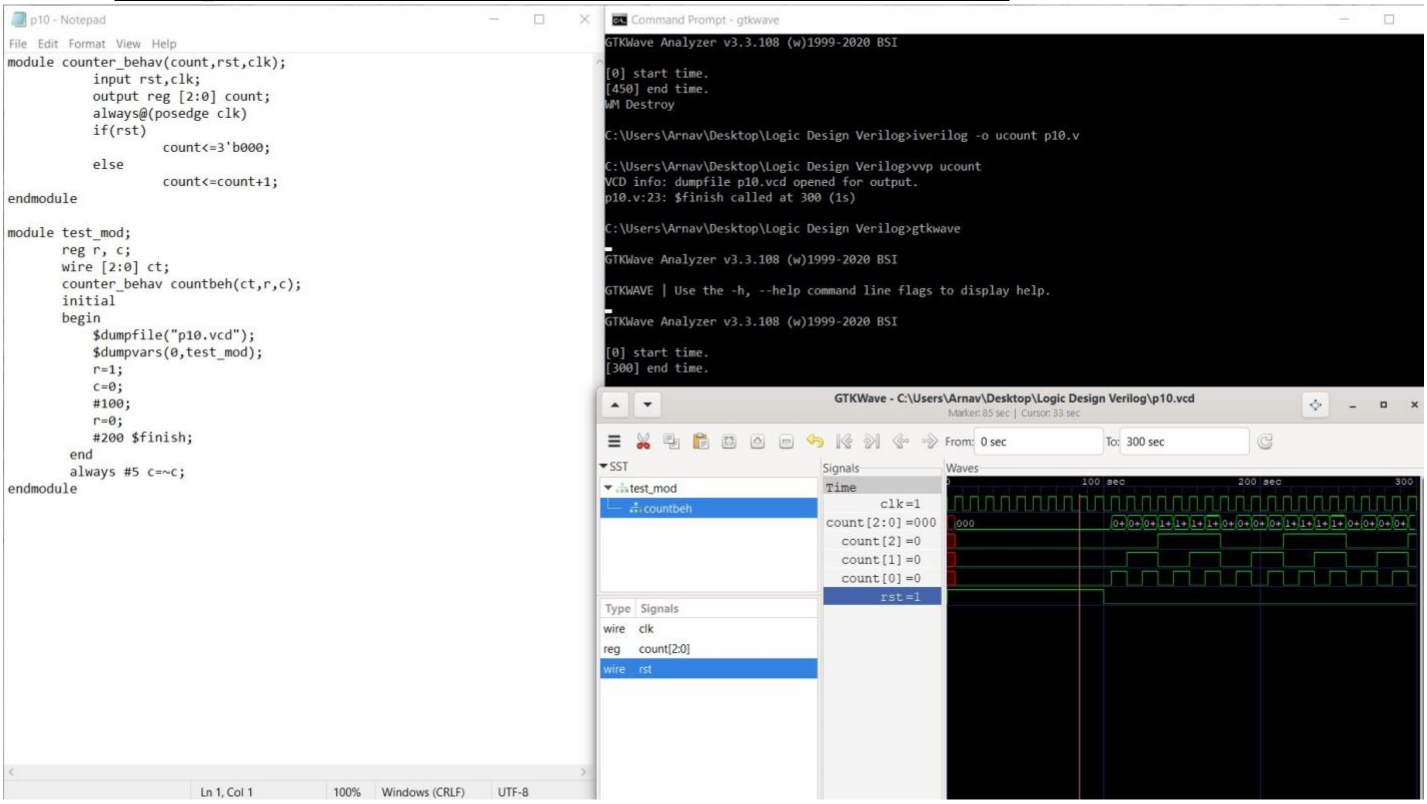
Experiment 10

module counter_behav(count,rst,clk); input rst,clk; output reg
[2:0] count; always@(posedge clk) //code executed for
every positive edge if(rst)
count<=3'b000;
else count<=count+1; //clock pulse count incremented
endmodule

module test_mod;
reg r,c; wire
[2:0] ct;
counter_behav countbeh(ct,r,c);
initial
begin
$dumpfile("cnt_test1.vcd");
$dumpvars(0,test_mod);
r=1; //for first 100ns reset r has value 1, clock c has value 0
c=0;

```
        #100;                  //for next 200ns reset r has value 0, clock c
        r=0;
        #200 $finish;
        end
        always #5 c=~c;
endmodule
```

Compilation, Execution and Result of Simulation



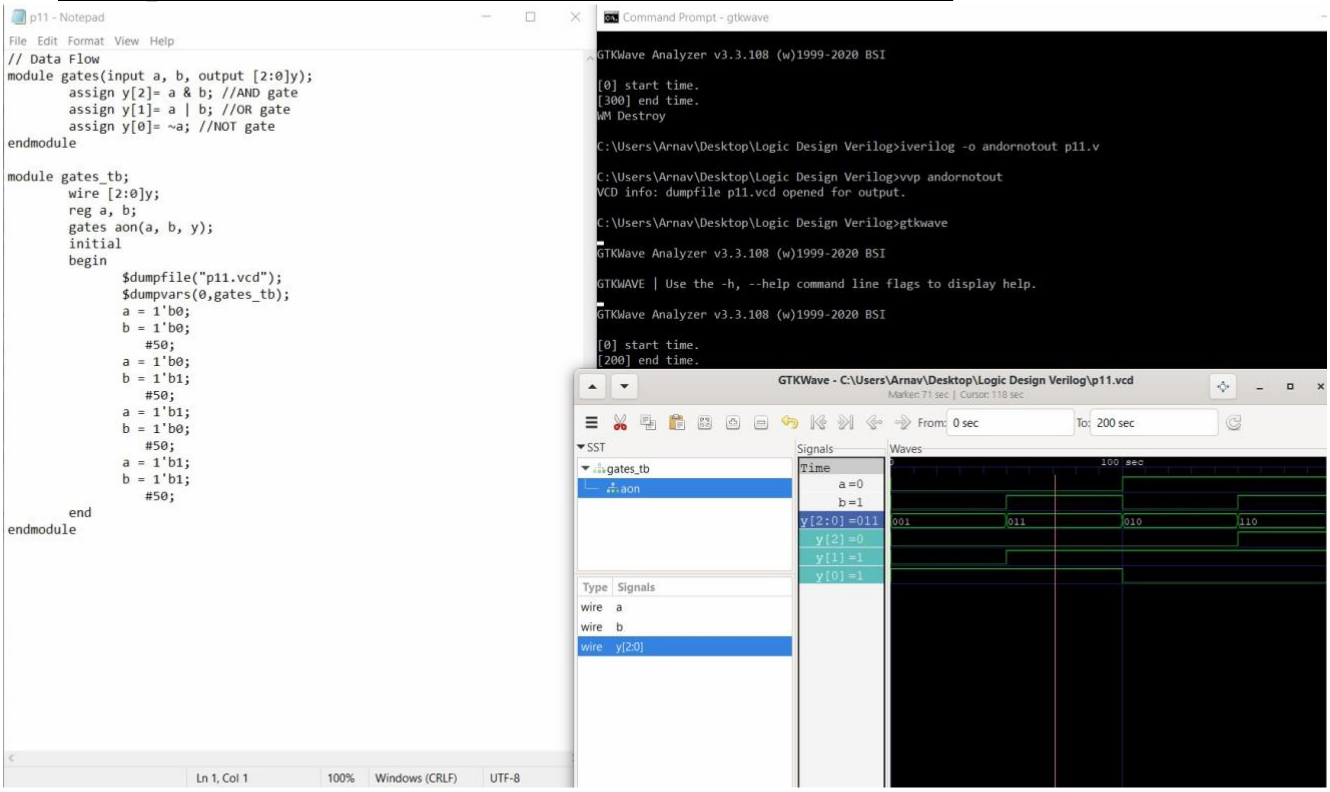
CYCLE 3: DATAFLOW MODELLING

Experiment 11

```
module gates(input a,b,output[2:0]y);   // Circuit is described by means of their
function.
        assign y[2]= a & b; //The first output y[0] is defined by 'AND' of a,b assign
        y[1]= a |b;   //The output y[1] is defined by 'OR' off a,b assign y[0]= ~a;
                //The output y[2] is defined by and NOT of a
endmodule

module gates_tb; wire
        [2:0]y; reg a, b;
        gates aon(a,b,y);
        initial begin
                $dumpfile("gates_test.vcd"); $dumpvars(0,gates_tb); a = 1'b0;
                        //for first 50ns inputs a,b have values 0,0(represented by b = 1'b0;
                single bit)
                  #50; a = 1'b0;     //for next 50ns inputs a,b have values 0,1(represented
                by
                b = 1'b1;                 single bit)
                  #50; a = 1'b1; //for next 50ns inputs a,b have values 1,0(represented by b
                = 1'b0;  single bit)
                  #50; a = 1'b1;     //for next 50ns inputs a,b have values 1,1(represented
                by
                b = 1'b1;                 single bit)
                  #50;
        end
endmodule
```

Compilation, Execution and Result of Simulation



Experiment 12

```
module adder(input a, input b,input cin, output cout);
        assign s = a^b^cin;
        assign cout= (a&b) |(a& cin) |(b & cin);
endmodule

module test1; reg a, b,cin;
        wire s , cout; adder FA
        (a,b,cin,cout); initial
        begin
                $dumpfile("FA1.vcd");
                $dumpvars(0,test1);
                a = 1;          //for first 5ns, a and b have values 1,1 and carry-in
                b = 1;  cin       has value 0
                = 0; #5
                a = 1;          //for next 5ns, a and b have values 1,1 and carry-in
                b = 1;    cin     has value 1
                = 1; #5
                a = 0;          //for next 5ns, a and b have values 0,1 and carry-in
```

b = 1; cin has value 0
= 0; #5
#10 $finish; //the state is continued for 10ns ore after which time
end duration is ended using $finish command
endmodule

Compilation, Execution and Result of Simulation